

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

The number a_2 is defined as the limit of the given code as $n \rightarrow \infty$. Inside the code,

notice the difference in exponents:

- Initial sum: $72^{-10^{24}}$
- Loop term: $72^{-(10^{23} \cdot k)}$

Let $T = 72^{-10^{23}}$. Then the initial sum is T^{10} and the loop subtracts T^k for $k \geq 3$. As $n \rightarrow \infty$, the function $f(n)$ approaches $T^{10} + \frac{1}{4} - \frac{T^3}{1-T}$, giving

$$a_2 = 2 \left(T^{10} + \frac{1}{4} - \frac{T^3}{1-T} \right) + \frac{1}{2} = 1 + 2T^{10} - \frac{2T^3}{1-T}.$$

Since T is extremely small and positive, $T^3 \gg T^{10}$, so $a_2 \approx 1 - 2T^3 < 1$.

With dosage rates $a_1 = 1$ and $a_2 < 1$, the fastest way to reach total dosage 1 is to use only the first therapy. The unique optimal minimizer is $(x_1, x_2) = (1, 0)$.

The infinity-norm distance from $(0.3, 0.7)$ to $(1, 0)$ is $\max(|0.3 - 1|, |0.7 - 0|) = 0.7$, which is **not** ≤ 0.4 .

(Note: The later revelation that $(0.6, 0.4)$ is at most 0.45 away is true, since its distance is 0.4.)